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Grad-505

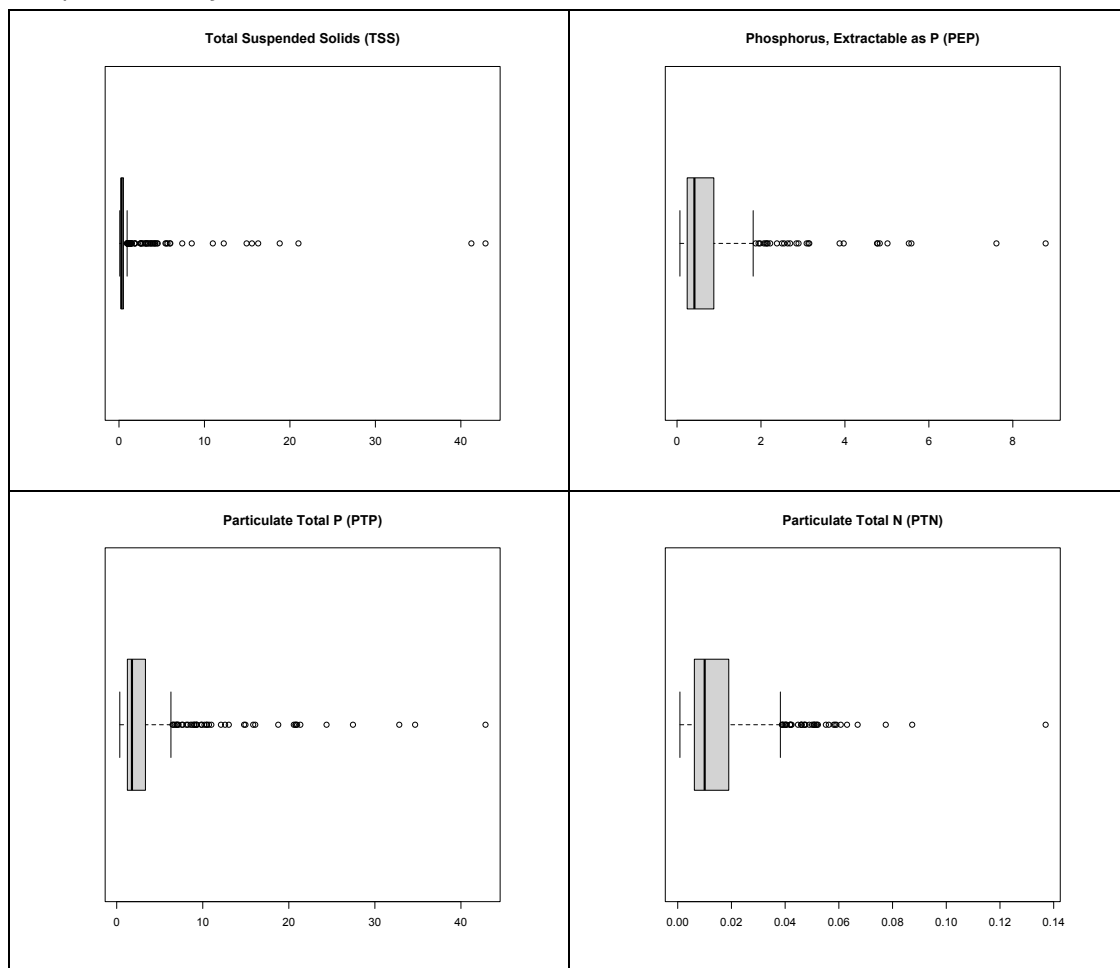
Project Deliverable 2

Exploratory Data Analysis

Identify any issues that exist with your dataset

I immediately see that all of my “qual” categorical variables have more than 80% missing data. In addition, one of my numeric response variables (phosphorus_extractable_p_ug_l or PEP) is missing 23% of its values. This level of incompleteness could reduce the sample size and add impactful bias to summary statistics.

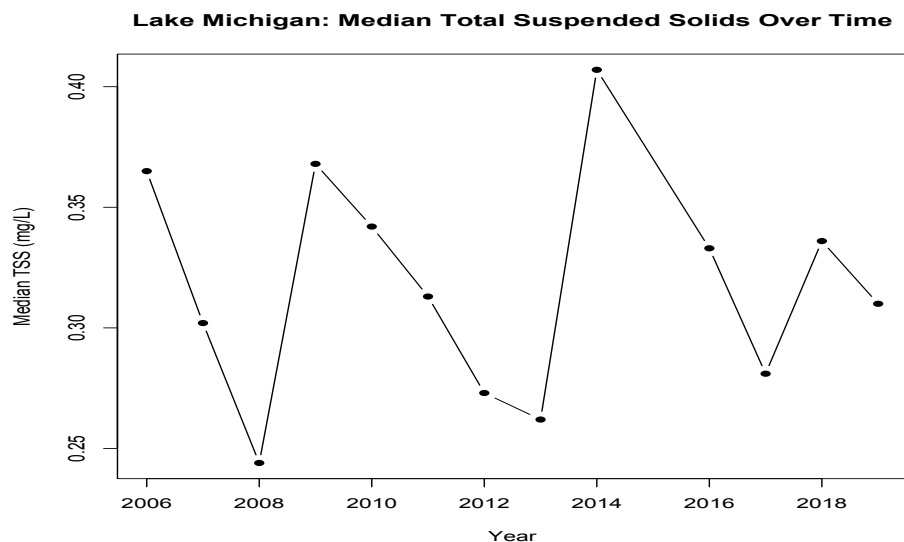
Further EDA using boxplots for four numeric response variables revealed that each is right-skewed, with longer upper whiskers, and appears to have dozens of upper outliers. This could suggest the presence of extreme values related to environmental instability or potential measurement errors. My observations of skewness and outliers suggest that my original assumptions and questions may not hold.



The questions, graphs, summary tables, and interpretations

1. **Regression** - How have nutrient particle levels changed over time in Lake Michigan? What predictions, or assumptions, can I make about today's levels based on past trends?

Interpretation of Time Series Visual (Median Lake Michigan – TSS levels)



SUMMARY TABLE

YEAR	MEDIAN	SIZE
01 2006	0.365	9
02 2007	0.302	9
03 2008	0.244	9
04 2009	0.368	9
05 2010	0.342	9
06 2011	0.313	9
07 2012	0.273	9
08 2013	0.262	9
09 2014	0.407	9
10 2016	0.333	9
11 2017	0.281	9
12 2018	0.336	9
13 2019	0.310	9

For preliminary EDA, I constructed a time-series plot showing the median levels of a single particulate nutrient in Lake Michigan year over year. Each point on the graph represents the median Total Suspended Solids (TSS) value for a given year, calculated from all available samples collected during that year. The median was used instead of the mean because the TSS data are right-skewed and contain outliers, which could make the mean less reliable as a representation of the population.

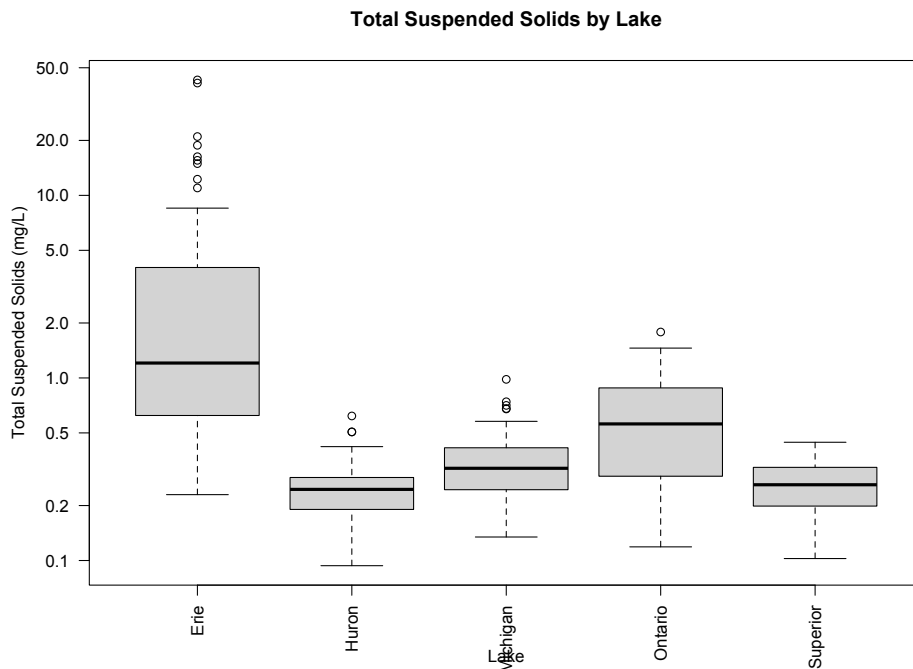
Overall, the plot shows noticeable variability over time. Some years show higher median concentrations than others, suggesting periods of increased particulate matter in the lake. This variability may reflect natural environmental fluctuations, such as changes in weather patterns, runoff, or seasonal conditions that influence sediment and nutrient levels (I'm thinking specifically of the 2014 polar vortex).

Gaps between 2014 and 2016 indicate no samples were available 2015, which limits the ability to draw strong conclusions about continuous trends. This gap further confirms that any predictions about current TSS levels based on past data should be made carefully.

In summary, the time series plot suggests that median TSS concentrations in Lake Michigan vary considerably from year to year, with no clear or stable long-term trend. Further analysis will be needed to support formal predictions or regression modeling.

2. Classification - Can I identify which lake a sample came from using combined particle collection and seasonal data?

Interpretation of Classification Attempt



LAKE	MEDIAN	IQR	SIZE
01 Erie:	1.207	3.404	83
02 Huron:	0.245	0.092	112
03 Michigan:	0.320	0.170	117
04 Ontario:	0.560	0.592	73
05 Superior:	0.260	0.125	115

For preliminary EDA, I created boxplots to examine total suspended solids (TSS) concentrations across all five Great Lakes. I used a log scale because the data is very skewed, and this makes it easier to see the extreme values. I wanted to see how the data was distributed, specifically the medians, IQRs, and outliers for each lake. The main goal was to determine whether TSS levels differ enough between lakes that I could use this data to predict which lake a sample came from.

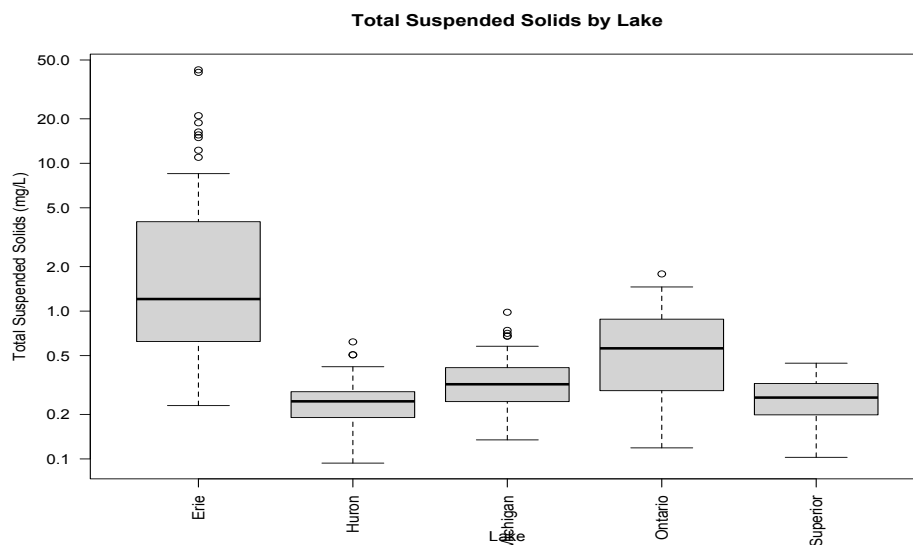
Looking at the plots, it's pretty clear that TSS levels aren't the same everywhere. Some lakes definitely have higher medians and wider spreads than others. They are all right-skewed and have a bunch of high-value outliers, but the medians and IQRs are different enough to suggest that the lakes have distinct, or at least somewhat distinct, characteristics.

The summary table supported this, showing that the median TSS and IQRs vary by lake. That makes me think the lake-specific characteristics really do impact particle levels. However, I noticed that some lakes have very similar distributions, which might make it tricky to classify them based on TSS alone.

Overall, it looks like TSS is useful for distinguishing the lakes, but the overlap between them means we probably can't achieve very high accuracy if we use only this one variable. Moving forward, I think I'll need to combine this with other nutrients or perhaps some seasonal data to improve classification results.

3. Anomaly Detection - Are there any outliers in the observations that suggest atypical environmental conditions for any of the lakes?

Interpretation of Anomaly Detection Attempt



LAKE	OUTLIERS	OBSERVATIONS
1Erie	9	83
2 Huron	3	112
3 Michigan	5	117
4 Ontario	1	73
5 Superior	0	115

Using the same boxplot from the previous question, it's easy to see the points sitting outside the upper whisker are potential outliers. Almost every lake has some interestingly high values, but they definitely vary in how many there are and how high they go. The extreme data points might mean TSS spikes more often in those specific locations. They could be real numbers representing environmental changes, like after a big storm or heavy snow, rather than just typos or broken sensors.

The summary table breaks down the number of outliers and the total samples for each lake. Even though it's a small percentage of the total data, these outliers can dramatically skew summary statistics or graphs if I'm not careful. The fact that the outlier counts are different across the lakes shows there's a lot of environmental variability going on.

In summary, preliminary anomaly detection shows that there are some really high TSS readings almost everywhere. I will take a closer look at them in later phases, but I think I should keep them in the dataset for now since they might be valid data points, not just noise.

Synopsis

This project examines nutrient particle data from the Great Lakes to see how levels change over time, how lakes differ from one another, and when unusually high or low values occur. Simple graphs and summaries help reveal trends, differences, and possible environmental events.

Keywords

#NutrientParticles, #GreatLakes, #OutlierDetection, #WaterQuality, #TimeSeriesTrends